ex. Given that $y_1 = e^{2t}$ is a solution to $y'' - 4y' + 4y = 0$, find a second linearly independent solution.

Char. eqn: $r^2 - 4r + 4 = 0$

$(r-2)^2 = 0$ $\quad r = 2$ is a double root

$y_1 = e^{2t}$, $y_2 = te^{2t}$ show why.

Reduction of order to prove form of 2nd soln.

$P(t) = -4$

$$e^{-\int P(t)dt} = e^{4t}$$

$$y_2(t) = y_1 \int \frac{e^{4t}}{y_1^2}\, dt$$

$$= e^{2t} \int \frac{e^{4t}}{e^{4t}}\, dt$$

$$= e^{2t} \int dt$$

$$= te^{2t}$$

ex. Given $y_1(t) = t$ is a solution to $y'' - \frac{1}{t}y' + \frac{1}{t^2}y = 0$.

$$y_2(t) = y_1(t) \int \frac{e^{-\int P(t)\,dt}}{(y_1(t))^2}\,dt$$

$$= t \int \frac{e^{-\int -\frac{1}{t}dt}}{t^2}\,dt$$

$$= t \int \frac{e^{\ln t}}{t^2}\,dt$$

$$= t \int \frac{t}{t^2}\,dt$$

$$= t \int \frac{dt}{t} = t \ln t$$

2. Solve the Cauchy-Euler equation $t^2y'' + 6ty + 6y = 0$

$a = 1 \quad b = 6 \quad c = 6$

$ar^2 + (b-a)r + c = 0$

$r^2 + 5r + 6 = 0$

$(r+3)(r+2) = 0$

$r = -2, -3$

$y = c_1 t^{-2} + c_2 t^{-3}$

3. Let $t^2y'' + 6ty' + 6y = 0, \quad t > 0$, with $y_1 = t^{-2}$. Find a second linearly independent solution using reduction of order.

$y'' + \frac{6}{t}y' + \frac{6}{t^2}y = 0$

$\int \frac{6}{t}dt = 6\ln t$

$P(t) = \frac{6}{t}$

$V = \int \frac{e^{-\int P(t)dt}}{y_1^2} = \int \frac{e^{-\int \frac{6}{t}dt}}{(t^{-2})^2} = \int \frac{t^{-6}}{t^{-4}}dt$

$= \int t^{-2}dt = -t^{-1}$

$y_2 = y_1 v = t^{-2}(-t^{-1}) = -t^{-3}$